LAB 03

Subject

Troubleshooting of SMPS (Switched Mode Power Supply)

Objective

- To identify common faults in a PC power supply (SMPS)
- To test and troubleshoot a faulty power supply
- To measure and verify output voltages of the SMPS
- To understand safety precautions while handling a power supply

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Equipment / Tools Required

- PC SMPS
- Multimeter
- Power cable
- Paper clip or jumper wire
- Screwdriver set
- Test load (motherboard or dummy load)
- Anti-static wrist strap

Theory

A **Switched Mode Power Supply (SMPS)** converts AC power from the mains into regulated DC voltages required by computer components.
Typical output voltages are:

- **+12V** (Yellow wire)
- **+5V** (Red wire)
- **+3.3V** (Orange wire)
- **Ground** (Black wire)

The SMPS includes protection circuits such as over-voltage, over-current, and short-circuit protection. Failure of any internal component may cause no power, unstable voltage, or shutdown.

Safety Precautions

1. Switch off and unplug the power supply before opening the cabinet.
2. Do not touch exposed components while power is ON.
3. Discharge capacitors before testing.
4. Use insulated tools only.
5. Avoid short-circuiting output terminals.

Procedure

Step 1: Visual Inspection

- Check power cable and connector.
- Look for burnt smell, bulged capacitors, or damaged wires.
- Ensure the cooling fan rotates freely.

Step 2: Power-On Test (Paper Clip Test)

1. Disconnect SMPS from motherboard.
2. Insert a paper clip between:
    - **Green wire (PS_ON)** and any **Black wire (Ground)** on the 24-pin connector.
3. Connect power cable and switch ON.
4. Observe whether the fan spins.

Observation:

- Fan spins → SMPS is receiving power
- Fan does not spin → SMPS may be faulty

Step 3: Voltage Testing Using Multimeter

1. Set multimeter to DC voltage.
2. Measure output voltages:
    - Yellow to Black → +12V
    - Red to Black → +5V
    - Orange to Black → +3.3V
3. Compare readings with standard values.

Step 4: Load Testing

- Connect SMPS to motherboard and peripherals.
- Power ON the system.
- Observe system stability and boot process.

Observations Table

Output Rail	Expected Voltage	Measured Voltage
+12V	11.4 – 12.6 V	_____
+5V	4.75 – 5.25 V	_____
+3.3V	3.14 – 3.47 V	_____

Result

The PC power supply was successfully tested and troubleshooted. Output voltages were found to be **within / not within** acceptable limits. The SMPS was determined to be **working / faulty**.

Conclusion

This lab helped in understanding the working of a PC SMPS and the practical methods used to troubleshoot power supply problems safely and effectively.